Product distribution of a VO_x/TiO_2 catalyst in methanol oxidation – influence of active sites and resulting coupled catalytic cycles

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Introduction

Dimethoxymethane (DMM), dimethyl ether (DME) and methyl formate (MF) are important platforms in the chemical industry. The heterogeneously catalyzed partial oxidation of green methanol on a bifunctional catalyst is a promising and sustainable alternative that improves the synthesis routes commonly used today. Vanadium oxide, supported on titanium dioxide (VO_x/TiO_2), enables the production of all three oxygenates (DMM, DME, MF) in a single process, which minimizes the demand for resources and energy [1,2]. To optimize the product spectra, it is important to investigate the reaction mechanism and the role of catalytic active sites present.

Experimental and proposed catalytic cycles

To study the influence of the active sites of the catalyst on product distribution, VO_x/TiO_2 catalysts were prepared and calcinated at different temperatures (300-600 °C) and characterized by TPD, TPO, TPR, SEM, BET, XRD, Raman spectroscopy and catalytic activity tests. This allowed to gain an insight into the interplay between the catalyst morphology including active sites and the resulting product spectra. The findings were compared to a proposed catalytic cycle (Fig. 1) [3]. The reactions involved are attributed to two main reaction pathways 1) redox reactions with oxygen and 2) dehydration reactions without oxygen. Redox sites promote the generation of intermediates and by-products, which are formaldehyde (FA), formic acid (FAc), carbon monoxide (CO) and carbon dioxide (CO_2) (Fig. 1). Brønsted sites catalyze the formation of DME and DMM (Fig. 1). MF is formed on a strong acid surface – the Lewis sites (Fig. 1). As recognized in the kinetic activity tests Fig. 2a-c illustrates the product spectrum shifts from MF to DMM and DME between T_{calc} = 500 and 600 °C, revealing a shift from Lewis to Brønsted sites. The XRD measurements confirm a phase shift from anatase to rutile TiO_2 at calcination temperatures of T_{calc} = 600 °C (Fig. 2 d)), which influences the morphology of the catalyst and consequently the active sites on the catalyst surface.

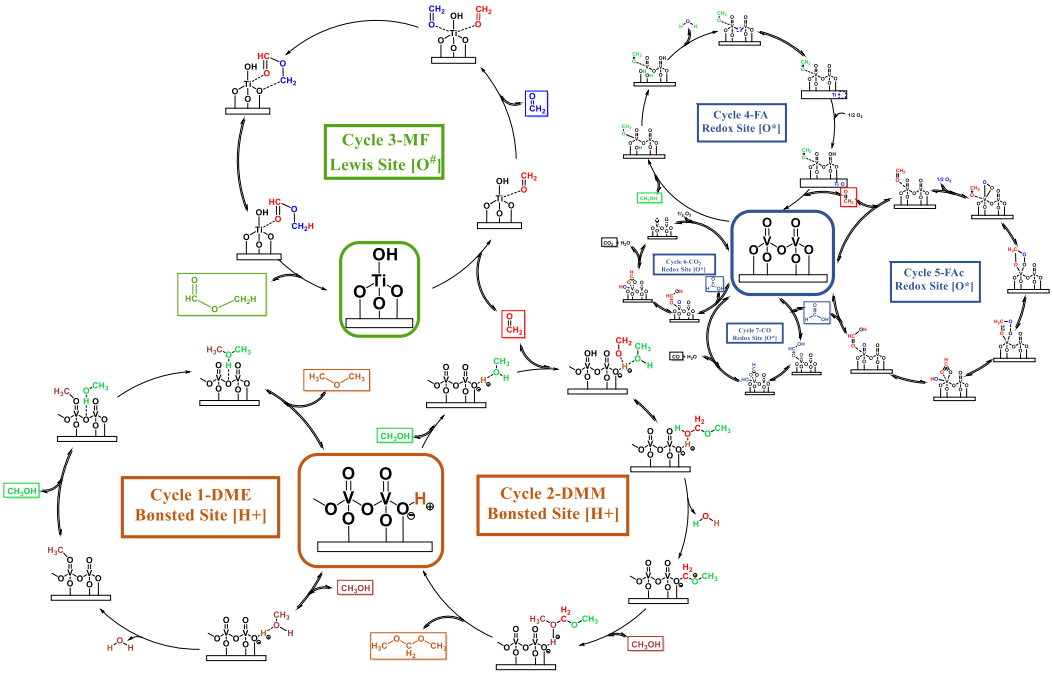

Figure 1: Catalytic cycles of the partial methanol oxidation over a VO_x/TiO_2 catalyst (based on [3])

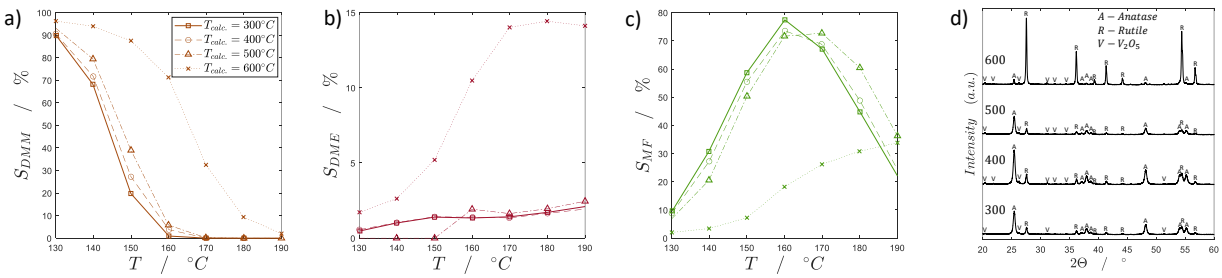

Figure 2: Selectivity of a) DMM, b) DME, c) MF and d) XRD measurements of the catalysts calcinated at different temperatures

Conclusion and Outlook

The results presented in this contribution reveal a significant relation between the catalyst morphology, influenced by the calcination temperature, and thus the resulting product spectra. To gain a deeper understanding of the catalytic cycles and the underlying kinetic mechanisms, perturbations with intermediates/products are performed, triggering reaction rates and (sub-)networks for analysis, respectively.

References

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